

TRACK: SMART MARITIME / AI / LOGISTICS

TEAM: DOCKNOVA DEVELOPMENT TEAM

DockNova

Smart Port Control Tower

Predict Port Congestion. Plan Better. Reduce Waiting.



The Problem: Port Congestion

- Ships wait a long time for a berth
- Port teams don't know where congestion will happen
- Manual planning takes too much time
- Waiting ships use fuel and create CO₂

\$15K–\$35K / Day

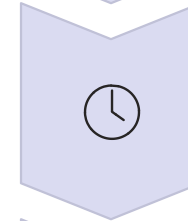
Possible cost

For one large ship waiting

Simple Flow



Ship Arrives



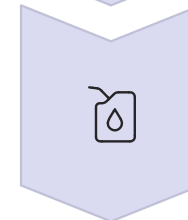
Berth Not Ready



Ship Waits

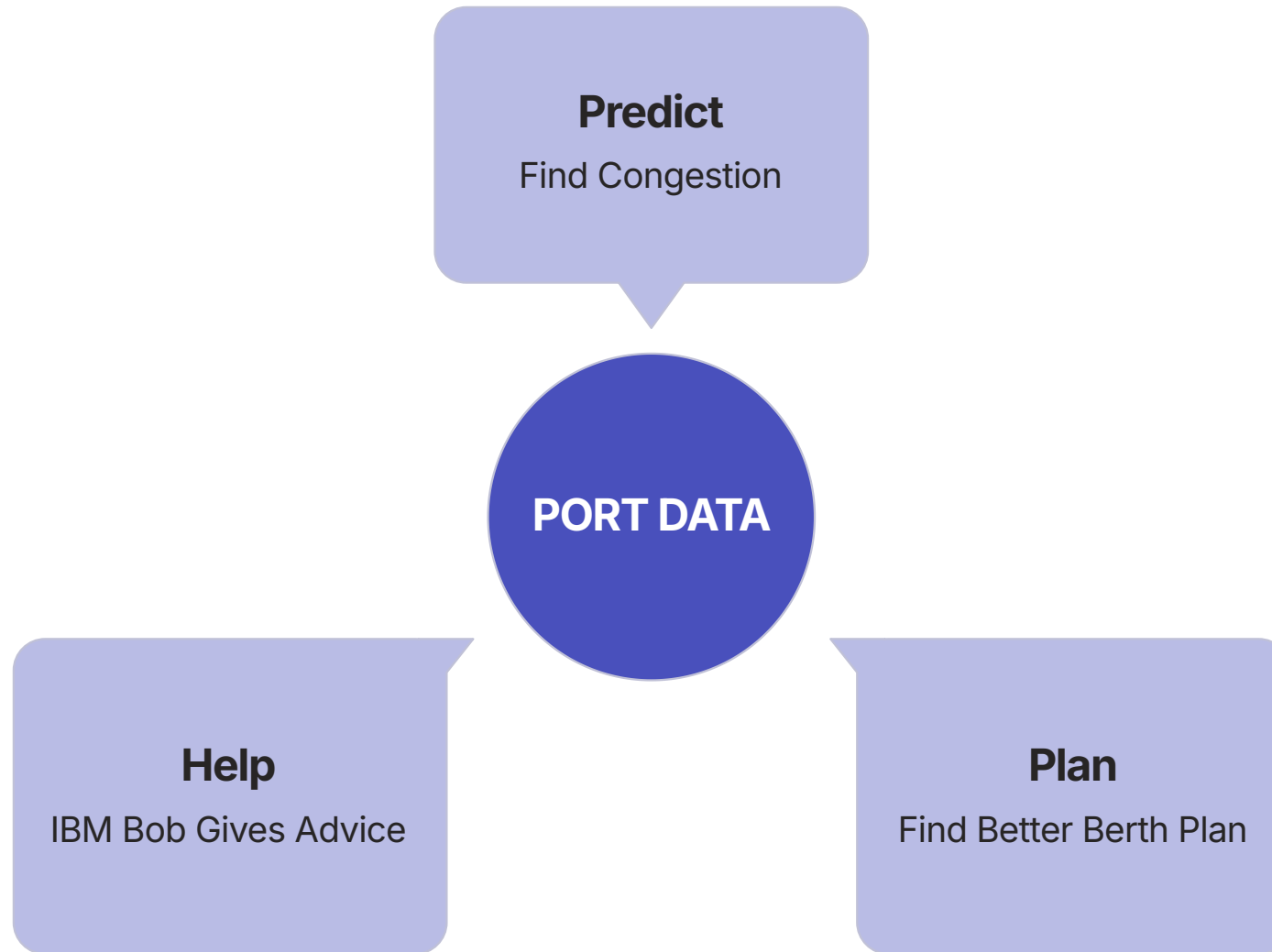


Port Gets Crowded



More Cost + Fuel + CO₂

Our Solution: DockNova



- **Predict:** Find possible congestion up to 72 hours ahead
- **Plan:** Find better berth and crane assignments
- **Help:** IBM Bob gives simple recommendations

How DockNova Works

Step 1: Collect Data

Ship, berth and port
information

Step 2: Predict

ML checks possible
congestion

Step 3: Plan

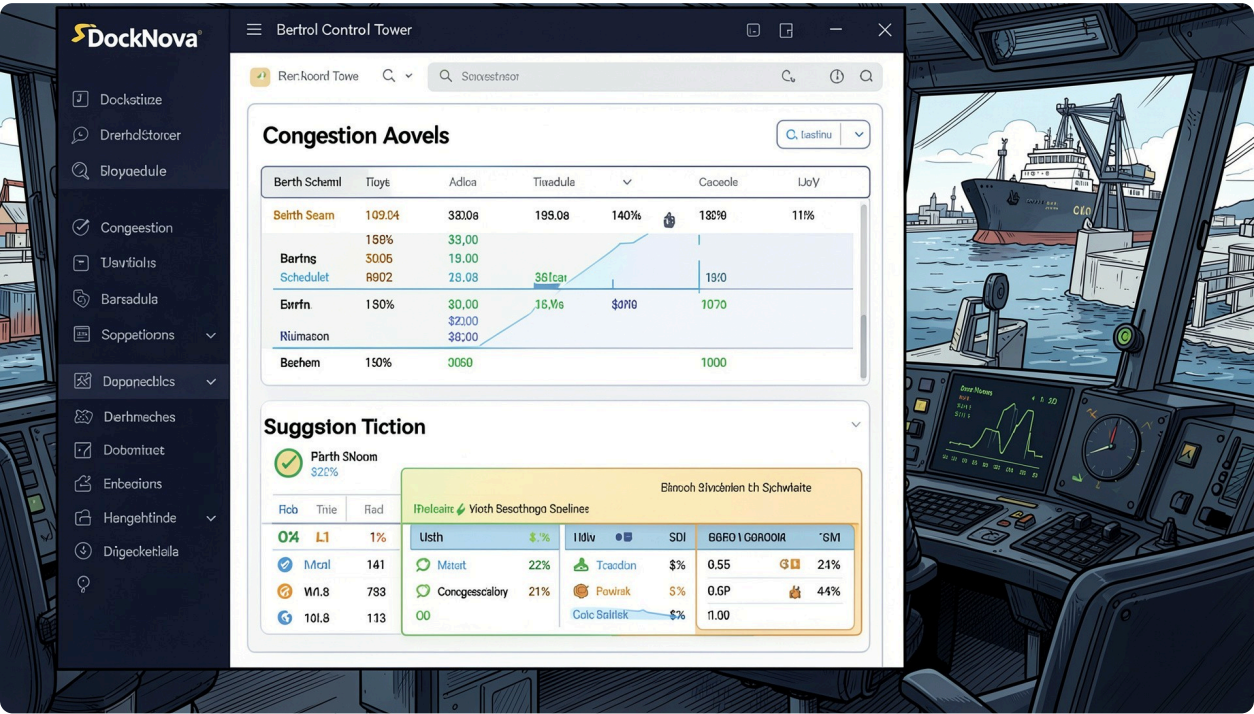
OR-Tools finds better
berth plan

Step 4: Suggest

IBM Bob explains and
gives advice

Data → Prediction → Planning → Action

DockNova Control Tower - Live Demo



Live Port Status

Vessel Information

72-Hour Prediction

Berth Planning

Port KPIs

Primary dashboard screenshot — readable UI for dispatch and decisioning

01

See Problem

03

Change Plan

02

Get Prediction

04

Take Action

IBM Bob AI Assistant

What does IBM Bob do?

IBM Bob works like an AI assistant for port operators.

Example:

Operator asks: "When will MV Ocean Star berth?"

IBM Bob gives:

- Expected berth time
- Reason for the delay
- Congestion information
- Suggested action

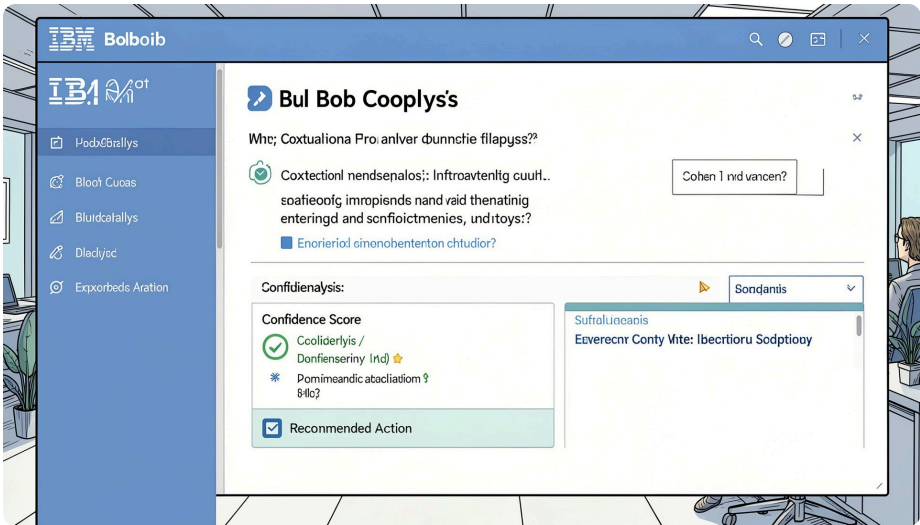
Operator Question

IBM Bob

Check Port Info

Simple Explanation

Suggested Action



IBM Bob — simple question and answer view

  SIMULATED DEMO

The current demo uses a simulated watsonx service.

What DockNova Can Improve

Less Waiting

Better berth planning can help reduce unnecessary waiting.

Less Cost

Less waiting can help reduce demurrage costs.

Less CO₂

Less vessel waiting can reduce unnecessary fuel use.

Better Decisions

Port teams get predictions and AI-based suggestions.

MAIN FEATURES:

- 72-Hour Prediction
- Berth Optimization
- AI Assistant
- Emergency Simulation

 Current demo uses simulated external data and AI services.

Team DockNova

Member 1

Name : Krina Patel

Frontend Developer

Work: Control Tower UI, Dashboard, Vercel Operator & System Admin interface

Member 2

Name : Bhumi Baldania

Frontend Developer

Work: AI Bob Interface, Charts, User Screens , Port Manager interface

Member 3

Name : Rishi Ramani

Backend Developer

Work: FastAPI Backend, ML Prediction, OR-Tools, APIs

Member 4

Name : Shalvi Patel

Documentation & Support

Work: Project Docs, PPT, Research, Testing, Demo Prep

DockNova

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Thank You — Questions?